The Model, Not the Search: Learning How Repertoires React to Damage for Few-Trial Recovery

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Abstract

Damage recovery with Intelligent Trial and Error (ITE) is a flagship application of Quality-Diversity (QD): after a damage, Bayesian optimisation searches a repertoire of gaits learned in simulation, using the intact performance as prior and the behaviour descriptor as similarity. Since QD relies on reinforcement learning (RL), adaptation is mostly evaluated with an oracle, and the number of trials on the damaged robot is no longer measured. We measure it on 33 repertoires of three simulated robots (Ant, Walker2d, HalfCheetah) built by MAP-Elites, PGA-ME and DCRL-ME. Repertoires built with RL are as resilient to damage as those of MAP-Elites and keep much better gaits, and adaptation needs a handful of trials where RL retraining needs hundreds of episodes. But ITE is barely better than trying the best intact gaits in order, even with a better descriptor: its model, not its search, is the bottleneck. Playing the repertoire under 64 simulated damages before deployment gives a prior and a kernel that turn ITE into a Bayesian linear model of the damage (ITE-SD). After 5 trials, ITE-SD improves on ITE by +0.079 in the share of the intact progress recovered on damages of the simulated family and by +0.072 under a reality gap, on every robot and QD algorithm (on Walker2d, by choosing gaits that do not fall rather than gaits that walk), without loss on unexpected damages. Reading each trial through proprioceptive signatures (ITE-Sig) adds another +0.034. A damage-robust TD3 policy is better than any adaptation on Ant within its training family, but its advantage disappears on unexpected damages.

Keywords

Quality-Diversity, damage recovery, Bayesian optimisation, Intelligent Trial and Error, MAP-Elites

1 Introduction

A robot that loses a motor in the field cannot be retrained from scratch. Animals keep walking after an injury because they already know many ways to move: they limp, shift their weight, use another leg more. Intelligent Trial and Error (ITE) [8] gave robots the same ability. Before deployment, MAP-Elites [17] builds in simulation a repertoire of diverse gaits; after a damage, Bayesian optimisation chooses which gaits to try on the robot, starting from their intact performance and assuming that gaits with similar behaviour descriptors react similarly (Figure 1). A damaged hexapod recovered in less than two minutes, and damage recovery has remained a flagship application of Quality-Diversity (QD).

Since then, QD algorithms have adopted reinforcement learning (RL) [9, 19] and accelerated simulators [6, 12]. They build much better gaits, but RL pushes towards performance, possibly at the cost of the diversity that adaptation needs, and adaptation is now mostly measured with an oracle that evaluates every policy of the repertoire after the perturbation [5]. The quantity that matters on hardware, the number of trials, is rarely reported. This work measures it, on three simulated robots and with three QD algorithms, and follows four questions.

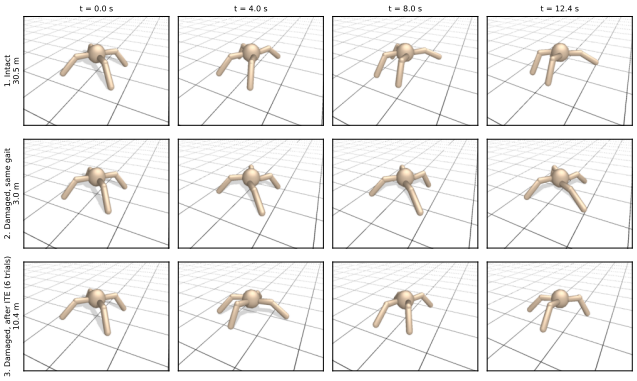


Figure 1: Damage recovery with a repertoire on the simulated Ant. Top: the best gait of a DCRL-ME repertoire. Middle: the same gait after the front-right leg is paralysed. Bottom: the gait found by ITE in six trials. Distances travelled in 12.5 s on the left.

(1) *Do modern repertoires survive damage?* Yes: repertoires built with RL are as resilient as those of MAP-Elites and keep much better gaits (Section 4).
(2) *Does the Bayesian search of ITE beat a simple ranking?* Barely. Trying the best intact gaits in order is almost as good, even with a better descriptor or a reality gap. We trace this to the *model* ITE is given, not to the search: its prior ignores which gaits tend to survive damages, and its kernel, a distance between descriptors chosen for diversity, says little about which gaits react alike (Section 5).
(3) *Can a better model be learned before deployment?* Yes. Playing the repertoire under a few dozen simulated damages gives a prior and a kernel that turn ITE into a Bayesian linear model of the damage (ITE-SD); reading the robot's proprioception during trials improves it further (ITE-Sig). Both beat ITE on every robot and QD algorithm for damages of the simulated family and under a reality gap, but not significantly for unexpected damages (Sections 6 and 7).
(4) *How does this compare with RL?* Re-training needs hundreds of episodes on the damaged robot. A damage-robust policy needs none and wins on Ant within its training family, but loses its advantage on unexpected damages (Section 8).

Contributions. (i) A replay protocol that makes adaptation methods cheap to compare at scale: 33 repertoires of three robots, every gait played under every test condition, hypotheses fixed before the final runs, and the replay validated against fresh episodes. (ii)

A diagnosis of ITE with modern repertoires. (iii) ITE-SD, which applies an empirical-Bayes GP prior [28] learned from simulated damages to the repertoire, and ITE-Sig, which also reads the damage in proprioception. (iv) An analysis of when a learned model helps, and a comparison, counted in episodes, with RL re-training and with a damage-robust policy.

2 Background and related work

Quality-Diversity and MAP-Elites. A QD algorithm searches for a *collection* of solutions that are both diverse and high-performing. MAP-Elites [17] divides a user-defined *behaviour descriptor* space into cells and keeps, in each cell, the best solution found (its *elite*). At each iteration, elites are selected at random, mutated, evaluated, and inserted in their cell if they beat its current occupant. For a walking robot, a solution is a policy (a neural network mapping sensors to motor commands), and the descriptor describes its gait, for instance how often each foot touches the ground. Evaluations are noisy, which makes elites look better than they are; this is the topic of uncertain QD [11].

Repertoire-based damage recovery. ITE [8] combines a MAP-Elites repertoire [17] with Bayesian optimisation over its cells, using the intact performance as prior. Follow-ups removed resets between trials [7], added safety constraints with simulated priors [20], selected among priors computed for several situations, either for direct policy search [21] or among several repertoires [15], meta-learned embeddings of simulated dynamics models [14], and used hierarchical repertoires on a physical hexapod [2]. Closest to our question, Bossens et al. [4] meta-evolve the behaviour space of the map so that it adapts better to damage, which requires a costly outer loop of repertoire building. We keep the repertoire fixed and only learn, cheaply, the model used to search it.

QD with reinforcement learning. PGA-ME [19] improves half of the offspring with the gradient of a TD3 critic [13]; DCRL-ME [9] conditions critic and actor on the descriptor. These methods, run in the QDax library [6] on Brax [12], produce much better gaits than MAP-Elites, and are evaluated on adaptation tasks by an oracle that tries every policy after the perturbation [5]: QD methods were more robust to large perturbations than skill-discovery RL, and PGA-ME adapted better than MAP-Elites, but the number of trials was not measured.

Learned similarity for Bayesian optimisation. Kernels learned from simulation make Bayesian optimisation of locomotion controllers more sample-efficient on hardware [3, 24], and priors meta-learned from related tasks speed up (safe) Bayesian optimisation [10, 25, 26], for instance through a Bayesian linear model on a basis shared across tasks [23]. Closest to ITE-SD, Wang et al. [28] estimate the mean and the covariance of a GP prior from functions observed offline on a finite set of candidates (an empirical-Bayes approach) and bound the regret of GP-UCB with this estimate. ITE-SD applies the same idea to QD repertoires: the candidates are the policies of the repertoire and the offline functions are their responses to simulated damages. Our contributions are the diagnosis that makes it necessary, its use for damage recovery, ITE-Sig, which adds proprioceptive observations of the same latent damage, and the evaluation.

Adaptation without a repertoire. Domain randomisation trains one policy that works under many conditions [22, 27]; other approaches identify the condition online from the recent history [16, 29] or meta-learn fast adaptation [18]. We compare with a damage-robust TD3 policy trained on the same damage family as our simulated damages, which needs no trial at all.

3 Experimental platform

All experiments share the same robots, repertoires, damages and measures. Two evaluation protocols are used: on Ant, adaptation is first run with fresh simulated episodes and compared with RL re-training (Sections 4, 5 and 8); then, to compare many methods on three robots, adaptation is replayed from stored episodes (Section 7).

3.1 Robots, task and repertoires

Robots. We use three Brax robots of QDax [6, 12] (Figure 2): Ant (`ant_uni`: a torso and four legs, each with a hip and an ankle motor, 8 actions, episodes of 250 steps or 12.5 s), Walker2d (`walker2d_uni`: two legs, 6 motors, 1000 steps or 8 s, the episode ends when the robot falls) and HalfCheetah (`halfcheetah_uni`: 6 motors, 12.5 s). The task is to run forward. The reward is the forward velocity minus control costs, plus, for Ant and Walker2d, a bonus for every step the robot stays alive (and a contact cost for Ant). For Ant, as in the DCRL-ME notebook, rewards are shifted by a constant and clipped at zero for all algorithms. The behaviour descriptor is the fraction of time each foot touches the ground ($[0, 1]^4$ for Ant, $[0, 1]^2$ for the others). For Ant, the mapping from actions to legs was checked in three independent ways (geometry at reset, motion of each foot when a single action is applied, and video): actions $(0, 1)$, $(2, 3)$, $(4, 5)$, $(6, 7)$ drive the hip and ankle of the front-left (FL), hind-left (HL), hind-right (HR) and front-right (FR) legs.

Repertoires. Each repertoire is a centroidal Voronoi tessellation of 1024 cells, built by MAP-Elites [17], PGA-ME [19] or DCRL-ME [9]. The three algorithms share the environment, the policy (MLP with two hidden layers of 128 units, tanh output), the repertoire and the number of episodes per iteration (256); their specific hyper-parameters are those of the official QDax notebooks (Table 5), and a test checks that the three configurations only differ by their emitter. Each run lasts 2000 iterations (512,256 episodes). We use 5 seeds per algorithm for Ant and 3 for the other robots, 33 repertoires in total (the first experiments on Ant use seeds 0 to 2). One short run per algorithm and robot (200 iterations) is a *development* repertoire: every hyper-parameter of the adaptation methods is chosen on these runs, which are never used for the results.

What "equal budget" means. The budget is equal in *episodes*, which is what a robot would pay, not in computation: per iteration, PGA-ME also makes 300 critic and 100 policy-gradient steps and DCRL-ME 3000 and 150, so that DCRL-ME makes about eight times more gradient steps than PGA-ME (82 against 52 minutes per Ant run on a T4, against 8 for MAP-Elites). The notebook hyper-parameters were not tuned for these tasks; the ranking between PGA-ME and DCRL-ME may depend on this choice.

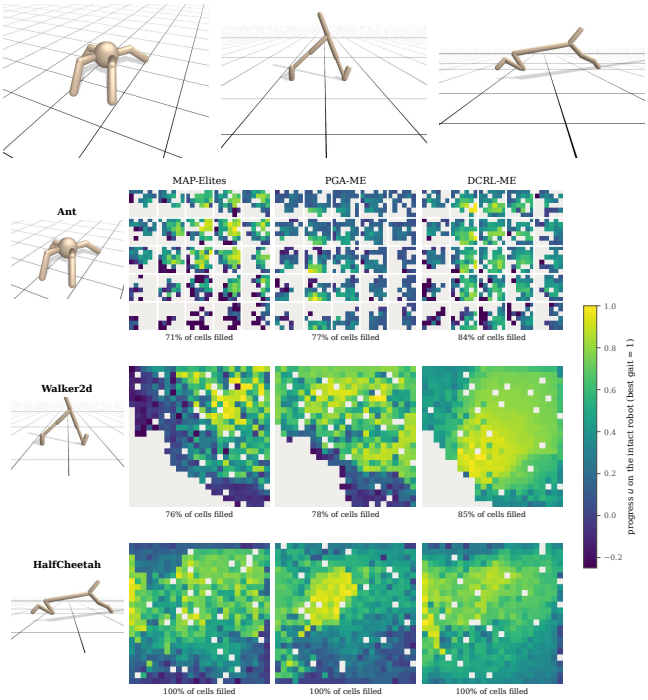


Figure 2: Top: the three robots (Ant, Walker2d, HalfCheetah) with the best gait of one repertoire. Bottom: one repertoire per robot and QD algorithm (seed 0). Each square is a gait, placed by how often each foot touches the ground (for Ant, feet 1–2 choose the large square and feet 3–4 the position inside it); colour is its progress on the intact robot.

3.2 Damages

A damage multiplies the commands of some motors by a factor λ ($\lambda = 0$: paralysed, $\lambda = 0.5$: weakened); since the torque of a motor is proportional to its command, this scales its strength, as in Chalumeau et al. [5]. Three families are used.

- *In-family damages*: each limb paralysed or weakened, and single joints paralysed. For Ant, the 12 scenarios are each leg paralysed, each leg weakened, two legs paralysed (FL+HL, same side; FL+FR, both front legs) and one ankle paralysed (FL, HR); 10 scenarios for the two other robots.
- *Out-of-family damages*, which act differently from a loss of strength: a jammed motor that keeps pushing (± 0.5), a miswired motor whose command is inverted, the joint sensors of a limb reading zero, a slippery foot (friction 0.05) and a foot four times heavier.
- A *reality gap*: the robot differs from the simulator that built the repertoire (ground friction $\times 0.7$, torso mass $\times 1.2$, motor strength $\times 0.9$; a second level at $\times 0.5$, $\times 1.4$, $\times 0.8$), alone or combined with each in-family damage.

The *simulated damages* used to learn a model (Section 6) and to train the damage-robust policy (Section 8) are drawn from the in-family distribution: each motor independently paralysed or weakened with probability 0.25; a draw equal to a test damage is redrawn.

3.3 Measures

Episodes are noisy (random initial state), and a robot that does not move still collects the survival bonus. We therefore measure performance from the return F_0 of a robot that plays no action. The main measure is the *share of the progress of the best intact gait* recovered by a method,

$$u = \frac{F - F_0}{F^{\star}_{\text{intact}} - F_0},$$

comparable across robots: $u = 0$ for a robot that plays no action, $u = 1$ for the best gait of the repertoire before the damage. For Ant, absolute performance is also reported as $v = (F - F_0)/S$ with $S = 1.2$ per step ($F_0 = 4.23$ per step), so that $v \approx 1$ for about 15 m in 12.5 s; this scale makes repertoires of different algorithms comparable. On Walker2d, a robot that plays no action falls, so a gait that stands without falling already has $u > 0$; we return to this in Section 7.3.

Prior and noisy elites. A repertoire stores the fitness of a single noisy episode, and MAP-Elites keeps the luckiest: elites are overestimated (on development repertoires the correlation between stored and re-evaluated fitness is 0.88, and the "best" cell of a MAP-Elites repertoire performs worse than a standing robot once re-evaluated). Since the repertoire lives in simulation, every gait is re-evaluated on the intact robot before deployment, and these values are the prior of all methods.

3.4 Evaluation protocols

Oracle and resilience. For each repertoire and damage, every gait is evaluated on 4 episodes. The best mean is the *oracle*, an upper bound for any method that picks a gait from the repertoire. The *resilience* of a repertoire to a damage is the share of the intact progress still reachable, $(F^{\star}_{\text{damaged}} - F_0)/(F^{\star}_{\text{intact}} - F_0)$. The oracle is the maximum of noisy means over several hundred cells and is biased upwards; the cell it selects is therefore re-evaluated on 16 fresh episodes (Section A).

Adaptation with fresh episodes (Ant). Each method runs for 20 trials on the damaged robot, one trial being one episode, and is repeated 5 times per (repertoire, damage). The quality of the gait it recommends (the best observed so far) is measured with its oracle mean, which the method never sees.

Replayed adaptation (three robots). To compare many methods on 33 repertoires, every gait of every repertoire is played under the intact robot (8 episodes), the 64 simulated damages (2 episodes each) and every test condition (8 episodes): 63 test conditions per Ant repertoire. Adaptation is then replayed from these *condition banks*: a trial returns one of the first four episodes of the chosen gait, drawn at random, and the recommended gait is scored with the mean of the other four, which no method sees. All methods share the same draws, and each is repeated 20 times per (repertoire, condition). Replayed ITE and top-k match their counterparts run with fresh episodes (Section 7.1).

Statistics. Repetitions of an adaptation only differ by episode noise and are averaged first: the statistical unit is a (repertoire, condition) pair. With fresh episodes on Ant, we report medians,

compare algorithms with two-sided Mann-Whitney tests and methods with paired two-sided Wilcoxon signed-rank tests. With replay, we report interquartile means (IQM) with 95 % stratified bootstrap intervals over repertoires [1], and paired Wilcoxon tests. Since units of one repertoire are not independent, every comparison is also checked with a linear mixed model with a random intercept per repertoire.

4 Do repertoires survive damage?

Adaptation can only find a good gait if the repertoire still contains one after the damage. We first check this, and how it depends on the QD algorithm, on Ant; we then look at how damages act on the repertoires of the three robots.

4.1 Repertoires on the intact robot

Both RL-based algorithms improve over MAP-Elites at equal budget, in different ways (Figure 3 and Table 6). DCRL-ME fills the most cells (83.8 % on Ant) and reaches the highest QD-score, consistent with its descriptor-conditioned critic improving gaits without changing their style. PGA-ME finds by far the fastest gaits (median best fitness 1922 against 1625 for DCRL-ME and 1235 for MAP-Elites): its unconditioned critic pushes offspring towards high reward, wherever they land in the grid. The fastest gait we inspected (PGA-ME) is a bounding gait in which the feet touch the ground only briefly. The same pattern holds on the other robots (Figure 2): DCRL-ME covers the most cells of Walker2d, and the two-feet descriptor of HalfCheetah is filled by all three algorithms.

4.2 Resilience to damage

In relative terms, the three kinds of repertoires are equally resilient: the median share of the intact progress that remains reachable after damage is 54 % for MAP-Elites, 73 % for PGA-ME and 64 % for DCRL-ME, and none of the pairwise differences is significant (Mann-Whitney, $p = 0.532$, $p = 0.996$ and $p = 0.496$; Figure 4). We find no sign that RL makes repertoires less useful after damage. In absolute terms the difference is large (Table 7): after damage, the best gait available is worth a median $v = 1.84$ in PGA-ME repertoires, $v = 1.14$ in DCRL-ME ones and only $v = 0.23$ in MAP-Elites ones. Weakening a leg is usually mild; paralysing a whole leg removes roughly half of the progress; with two paralysed legs, no repertoire keeps a useful gait. Correcting the selection bias of the oracle does not change these conclusions (Table 12).

4.3 How damages act on a repertoire

The condition banks show how every gait of the 33 repertoires reacts to every condition. Three facts shape what an adaptation method can do. First, the intact ranking is only a coarse guide: the mean rank correlation between intact and damaged performance is 0.44 for in-family damages and 0.50 out of the family, and the best gait after the damage lies outside the 10 best intact gaits in 86 % of the in-family units. Second, the responses of a repertoire to damages are low-dimensional (Figure 5, left): the first 4 principal components of the responses to the 64 simulated damages explain 73 % of their variance for Ant, 77 % for Walker2d and 75 % for HalfCheetah. Third, the response to a test damage is largely contained in the span of the simulated responses: on average 0.83 of its variance in the family, 0.78 out of it and 0.78 under a reality gap. A few trials could therefore identify a damage, if the adaptation method knew these directions; Section 5 shows that ITE does not.

Table 1: Ant, fresh episodes: adaptation methods (median over repertoires, damages and repetitions). "% intact": share of the progress of the best intact gait; "% oracle": share of the best gait available after damage.

Method	Trials	% intact	% oracle	Absolute v
ITE	5	33	62	0.37
top-k (no GP)	20	35	77	0.40
random	20	33	60	0.31
best-intact	1	18	34	0.19

5 Does Bayesian search beat a ranking?

5.1 Intelligent Trial and Error

ITE [8] places a Gaussian process (GP) on the performance of the gaits after damage. Its prior mean is the intact performance p_i of each gait and its kernel a Matérn-5/2 function of the distance between descriptors, $k(i,j) = \sigma_f^2\, \kappa_{5/2}(\|\mathbf{d}_i - \mathbf{d}_j\|/\ell)$. After t trials,

$$\mu_t(\mathbf{d}) = p(\mathbf{d}) + \mathbf{k}(\mathbf{d})^\top (K + \sigma_n^2 I)^{-1}\big(\mathbf{y} - p(\mathbf{D}_t)\big),$$
$$\sigma_t^2(\mathbf{d}) = \sigma_f^2 - \mathbf{k}(\mathbf{d})^\top (K + \sigma_n^2 I)^{-1}\mathbf{k}(\mathbf{d}),$$

where $\mathbf{D}_t$ and $\mathbf{y}$ are the descriptors and observed performances of the gaits tried so far. The next gait maximises $\mu_t + \beta\sigma_t$, and ITE stops when the best observed performance reaches $\alpha \max \mu_t$. Two assumptions are built in: the intact ranking is a good starting point, and gaits that are close in *descriptor* space react similarly to the damage.

Hyper-parameters. With the values of Cully et al. [8] (lengthscale 0.4, unit signal variance, noise 10^{-3}), ITE stopped after very few trials and plateaued: the residuals have a variance of 0.03–0.06, not 1, and a large nugget, because neighbouring cells react differently. We therefore fit $(\ell, \sigma_f^2, \sigma_n^2)$ by maximum marginal likelihood on the development repertoires (on Ant: $\ell = 0.4$, $\sigma_f^2 = 0.05$, $\sigma_n^2 = 0.02$, log-likelihood +4059 against −1280 for the original values; per robot in the replayed experiments). β and $\alpha = 0.9$ follow Cully et al. [8]; β made little difference on the development set.

Baselines. Best intact replays the best intact gait (one trial). *Top-k* tries the gaits in decreasing order of their intact performance, without any model. *Random* tries gaits at random.

5.2 ITE on Ant: little more than a ranking

ITE works. It meets its stopping criterion in 89 % of the runs, after a median of 5 trials (IQR 2–10), and the recommended gait recovers 33 % of the progress of the best intact gait, against 18 % when the best intact gait is simply replayed, and 62 % of what an exhaustive search of the repertoire would find (Table 1). Random search is much slower (9 % after three trials, against 28 %). Figure 7 shows one run.

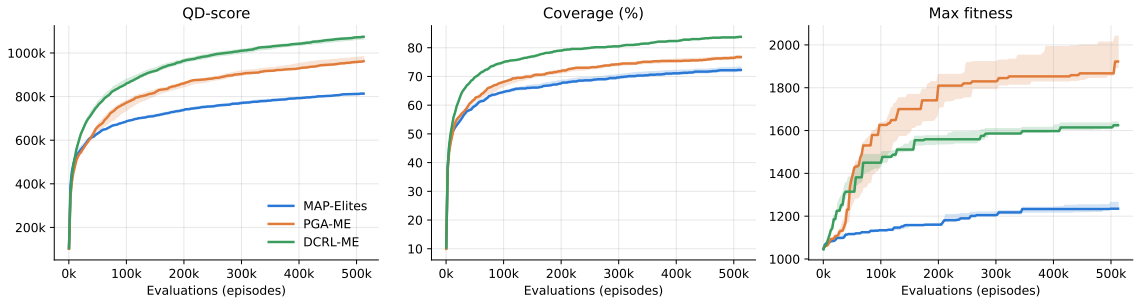


Figure 3: Building the Ant repertoires: QD-score, coverage and best fitness during training (median and inter-quartile range over 3 seeds).

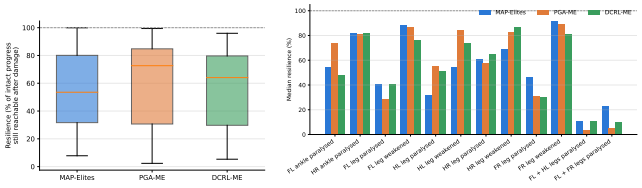


Figure 4: Resilience of the Ant repertoires: share of the intact oracle progress still reachable after damage, for all damages (left) and by damage (right, median over seeds).

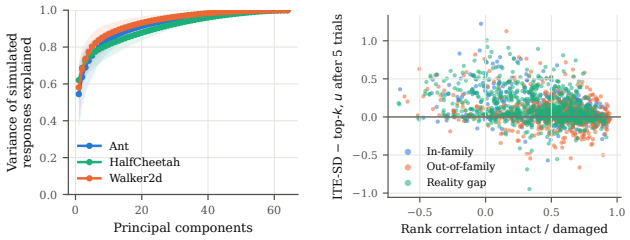


Figure 5: Left: variance of the responses of a repertoire to the 64 simulated damages explained by their first principal components (mean and range over repertoires). Right: gain of ITE-SD (Section 6) over top-k after 5 trials against the rank correlation between intact and damaged performance, one dot per (repertoire, condition) unit.

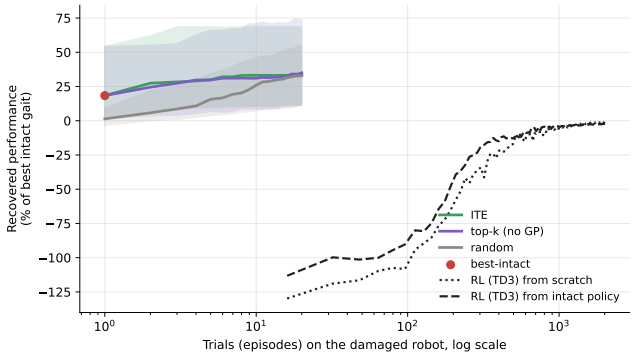


Figure 6: Ant, fresh episodes: performance of the recommended gait against the number of trials on the damaged robot (median and IQR over repertoires, damages and repetitions; each repertoire relative to its own best intact gait). RL curves (Section 8) are relative to the best intact DCRL-ME gait.

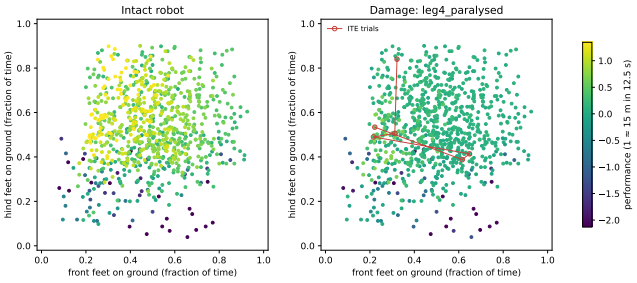


Figure 7: An Ant DCRL-ME repertoire before and after the front-right leg is paralysed (front against hind feet contact), and the six gaits tried by ITE.

The comparison with top-k is less favourable (Figure 6): after three trials ITE reaches 28 % and top-k 27 %; after 20 trials, 33 % and 35 %. Paired over the 108 (repertoire, damage) units, ITE is better than top-k after five trials in 47 % of them, equal in 5 % and worse in 48 % (Wilcoxon $p = 0.445$). After three trials ITE has a small advantage (mean difference +0.05 in v, $p = 0.009$), which comes mostly from DCRL-ME repertoires (+0.13, $p = 0.019$) and disappears after five and ten trials (Table 9). Mixed models find a positive mean gain on DCRL-ME after five trials (+0.08 [+0.02, +0.15], $p = 0.014$) while the median gain is zero: the model helps a lot on a few damages and not at all on the others. What a robot finally gets depends more on how its repertoire was built than on the adaptation method: with ITE, the recommended gait is worth $v = 1.47$ with PGA-ME, $v = 0.59$ with DCRL-ME and $v = 0.11$ with MAP-Elites (Table 11; at the level of seeds, every PGA-ME repertoire is above every DCRL-ME one, itself above every MAP-Elites one).

5.3 Neither a better descriptor nor a reality gap changes this

A better descriptor. With the feet-contact descriptor, the fitted noise variance (0.020) is not small compared with the signal variance (0.050): neighbouring cells react differently to the same damage. A damage acts on joints, so we built a second descriptor, the mean absolute command of each of the 8 motors on the intact robot, and fitted the GP on it in the same way (Table 15). It explains the residuals much better: the log-likelihood gained over independent cells is 152.2 per (repertoire, damage) pair against 51.7, and the fitted noise drops to 0.003. Yet ITE barely improves (Table 16): against ITE on feet contact, the mean gain is +0.02 after five trials ($p = 0.923$) and +0.03 after ten ($p = 0.757$). Against top-k, ITE on joint usage is ahead after three trials (+0.07, $p = 0.013$) but not after ten.

A reality gap. With fresh episodes, prior and trials come from the same physics, the easy case. To mimic a deployment, the trials were also run on a robot whose physics differ from the simulator (first level of Section 3.2, fixed before the experiment). The gap alone costs little: the best intact gait keeps 91 % of its progress. With a damage on top of it (Table 17), ITE finds $v = 0.28$ after a median of 6 trials, top-k reaches 0.27 after 20 trials and the best intact gait 0.10. ITE's early lead is slightly larger than without the gap (+0.06 after three trials, $p = 0.002$), but gone after five (+0.03, $p = 0.390$).

Searching a continuous space. DCRL-ME also learns an actor that should produce a gait for any requested descriptor. Letting ITE search 2048 such gaits instead of the repertoire cells is clearly worse: the actor's gaits are much weaker than the elites (best available gait after damage $v = 0.32$ against $v = 0.90$), and ITE on the actor finds $v = 0.14$ against $v = 0.53$ on the cells (Figure 13).

5.4 Three robots: ITE retries gaits, and its kernel fails on Walker2d

The replayed experiments (Section 3.4) confirm the picture on three robots and add two observations (Table 2). First, ITE tries the same cells again: forbidding repeats gives a small gain after 10 trials, not significant after correction (+0.009, $p = 0.074$), and a clearer one after 20 trials (+0.022, $p = 0.001$), on Ant ($p < 0.001$) but not significantly on Walker2d ($p = 0.071$) or HalfCheetah ($p = 0.714$). Second, the descriptor kernel can fail completely: on Walker2d, ITE recovers $u = 0.11$ after 5 trials, barely more than top-k (0.02), because foot contacts say nothing about which gaits keep the robot upright after a damage (Table 3).

5.5 Diagnosis: the model, not the search

ITE helps only if trying one gait tells something about others, and if the best gait after damage is not among the first ones a ranking would try. With the feet-contact descriptor, the first condition is weak: neighbouring cells react differently to the same damage. The joint-usage descriptor fixes this, yet ITE still gains little: the second condition fails too, because a damaged repertoire mostly keeps some of its best gaits and a re-evaluated ranking reaches them in a few trials. But Section 4.3 shows that this is not the whole story. Across robots, the best gait after a damage is often far down the intact ranking, and damages move the repertoire along a few directions that simulation can reveal. A Bayesian search should therefore help, provided that its prior knows which gaits tend to survive damages and its kernel knows which gaits react alike. Neither can be read from a descriptor chosen for diversity, but both can be learned in simulation.

6 Learning the adaptation model from simulated damages

We keep the search of ITE and replace its model. Both the prior and the kernel are learned, before deployment, from the way the repertoire reacts to simulated damages.

Setting. A repertoire $\mathcal{R} = \{\pi_1, \dots, \pi_n\}$ of n policies (gaits) has been built in simulation by a QD algorithm, each with a behaviour descriptor $\mathbf{d}_i$. After an unknown damage, the robot can play *trials*: one episode with a chosen policy, which returns a noisy measure of its performance. The goal is to find a good policy in as few trials as possible, measured by the share u of the intact progress that the recommended policy recovers (Section 3.3).

Simulated damage responses. Before any damage, the simulator that built the repertoire can also break it. We draw N damages $\delta_1, \dots, \delta_N$ from a family of plausible damages (here: each joint independently paralysed or weakened with probability 0.25, the family used to train damage-robust RL policies) and play every gait of the repertoire under each of them. This gives a response matrix

$$R \in \mathbb{R}^{N \times n}, \qquad R_{ki} = u_i(\delta_k) - p_i,$$

the change of performance of gait i under damage δ_k. It costs $2Nn$ simulated episodes, a few minutes on one GPU for $N = 64$ and $n \approx 1000$; no trial on the robot is involved.

ITE-SD. We keep the GP and the acquisition of ITE and change only its prior mean and kernel, which are now estimated from R:

$$m_i = p_i + \bar{r}_i, \qquad \bar{r} = \tfrac{1}{N} \textstyle\sum_k R_{k\cdot}, \tag{1}$$

$$k_{\mathrm{SD}}(i,j) = \tfrac{1}{N-1} \textstyle\sum_k (R_{ki} - \bar{r}_i)(R_{kj} - \bar{r}_j) + \tau^2 \mathbb{1}[i = j]. \tag{2}$$

The prior mean is the expected performance under a random damage; the kernel says that two gaits are similar when *they react in the same way to damages*, whatever their descriptors. The nugget τ^2 keeps some independent uncertainty per gait. A gait is never tried twice, and the recommended gait is the best one observed. This is the empirical-Bayes estimate of a GP prior over a finite set of candidates studied by Wang et al. [28], applied to the gaits of a repertoire.

Interpretation. Writing the centred responses as $U = (R - \bar{r})^\top / \sqrt{N-1} \in \mathbb{R}^{n \times N}$, the GP of ITE-SD is the Bayesian linear model

$$u(\delta) = m + U\mathbf{z} + \boldsymbol{\epsilon}, \qquad \mathbf{z} \sim \mathcal{N}(0, I_N), \ \boldsymbol{\epsilon} \sim \mathcal{N}(0, \tau^2 I_n).$$

A trial is thus a measurement that identifies the damage in the space spanned by the simulated responses, and the repertoire itself is the probe. If the effect of a damage on the repertoire is low-dimensional, as we find in Section 4.3, a few trials pin down $\mathbf{z}$ and predict all gaits. Choosing among N simulated priors [15] is the special case where $\mathbf{z}$ is restricted to the vertices of the simplex; ITE-SD can combine responses, for instance to represent a two-leg damage from one-leg damages. For damages unlike the simulated ones, we also test a hybrid kernel $k_{\mathrm{SD}} + k_{\mathrm{Matérn}}$.

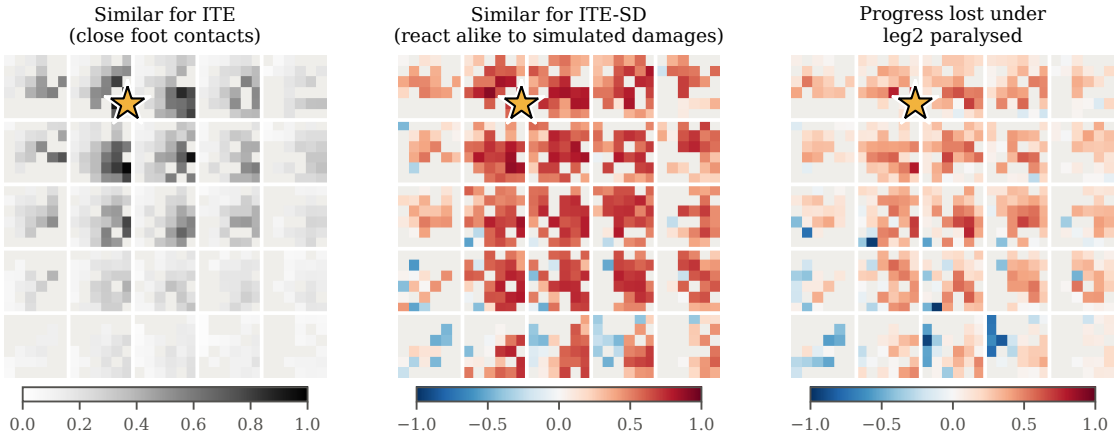


Figure 8: What each model considers similar to the best intact gait (star) of an Ant DCRL-ME repertoire. Left: the Matérn kernel of ITE, a smooth neighbourhood in descriptor space. Middle: the correlation learned by ITE-SD from 64 simulated damages, which links gaits across the whole repertoire. Right: the progress each gait actually loses when leg 2 is paralysed (red: loses). The star is the same gait in the three panels; the red regions of the middle and right panels match, the neighbourhood on the left does not.

ITE-Sig: reading the damage in proprioception. A trial returns more than a performance: during the episode the robot measures its joint positions and velocities, the motion of its torso and its foot contacts. We summarise an episode by a *signature* $\mathbf{s} \in \mathbb{R}^d$ (mean and mean absolute value of each proprioceptive channel, and the contact fraction of each foot), which a real robot gets from its encoders, IMU and contact sensors. The simulated damages already provide, for every gait, the signature it produces under each damage. ITE-Sig extends the linear model of ITE-SD so that the same latent damage $\mathbf{z}$ explains both:

$$u_i = m_i + U_i\mathbf{z} + \epsilon, \qquad \tilde{\mathbf{s}}_i = \mathbf{c}_i + V_i\mathbf{z} + \boldsymbol{\epsilon}', \quad \boldsymbol{\epsilon}' \sim \mathcal{N}(0, I_K/w),$$

where $\tilde{\mathbf{s}}$ is the signature whitened with the episode-to-episode noise of the intact robot and projected on the K directions along which simulated damages differ most relative to that noise, and $\mathbf{c}_i$, V_i are the mean and centred responses of these projected signatures. The posterior on $\mathbf{z}$ stays Gaussian and is updated in closed form; the acquisition and the recommendation are those of ITE-SD, and $w = 0$ gives back ITE-SD. Each trial thus informs $\mathbf{z}$ through $1+K$ numbers instead of one. A damage unlike the simulated ones could make the signatures misleading, so ITE-Sig also tests, after each trial, whether the observed signatures are plausible under the model (sum of squared Mahalanobis residuals against its χ^2 quantile at level p) and falls back to ITE-SD if not. K, w and p are chosen on the development repertoires. Inferring a latent description of the dynamics from proprioception is also the principle of RMA [16], which feeds it to a single policy; here it is a Gaussian posterior over the damage, shared with the performance model, that guides the choice among the gaits of the repertoire.

7 Evaluating the learned model

Methods compared. All methods search the same repertoires with the replay protocol of Section 3.4, for up to 20 trials. *Best intact*, *random* and *top-k* are the baselines of Section 5. *ITE* uses the Matérn kernel on the descriptor with GP hyper-parameters fitted per robot on the development repertoires; *ITE, no repeat* forbids trying a cell twice; *ITE, joint usage* uses the joint-usage descriptor. Three methods also use the simulated damages: *top-k on the simulated prior* ranks cells by m_i; *APROL-like* adapts the prior selection of Kaushik et al. [15]: each simulated damage provides a prior, and after every trial ITE uses the one with the highest GP marginal likelihood; ITE-SD; and ITE-Sig. Settings chosen on the development repertoires: $\beta = 0.3$ for ITE; $\beta = 0.3$, $\tau^2 = 0.005$ and noise 0.005 for ITE-SD and ITE-Sig; $K = 32$, $w = 0.1$ and $p = 0.01$ for ITE-Sig. The damage-robust TD3 policy of Section 8 plays without any trial.

Hypotheses. The following hypotheses were written before the final runs, with ITE-SD as the proposed method; their p-values are Holm-corrected together. **H1** (diagnosis): ITE wastes trials by trying the same gaits again; forbidding repeats improves it at 10 and 20 trials (Section 5). **H2**: on in-family damages, ITE-SD beats ITE and top-k after 3, 5 and 10 trials. **H3** (generalisation): on out-of-family damages and under a reality gap, ITE-SD remains at least as good as ITE, while the damage-robust policy degrades. **H4** (explanation): the gain of a model over top-k grows when the intact prior is a poor guide (low rank correlation between intact and damaged performance) and when the response lies in the span of the simulated responses. **H5** (generality): H1 and H2 hold for every robot and every QD algorithm. ITE-Sig was designed afterwards, on the development repertoires; its comparisons form a separate, exploratory family, also Holm-corrected.

7.1 Validity of the replay

On the 9 Ant repertoires of Section 5, ITE and top-k had been run with fresh simulated episodes. Replayed from the banks, the same methods give the same performance per (repertoire, damage) pair: correlation between 0.97 and 0.99 after 3 to 20 trials, and mean

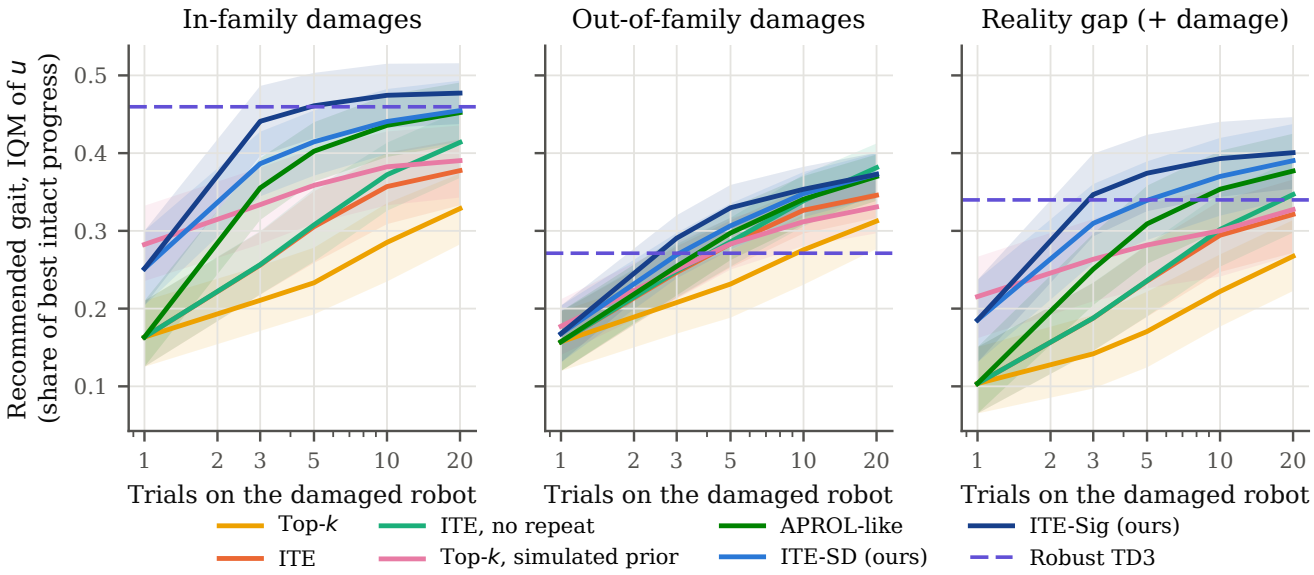


Figure 9: Performance of the recommended gait after k trials on the damaged robot (IQM over the (repertoire, condition) units of the three robots, 95 % stratified bootstrap interval). The dashed line is the damage-robust TD3 policy, which does not adapt (pooled over robots, including Walker2d where it failed to learn a gait; per-robot values in Table 3).

Table 2: IQM of u after 1, 5 and 20 trials, all robots pooled (33 repertoires; 360, 576 and 786 units). Best value of each column in bold. The robust TD3 policy plays without trials.

	In-family damages			Out-of-family damages			Reality gap (+ damage)		
Trials	1	5	20	1	5	20	1	5	20
Best intact	0.16	0.16	0.16	0.16	0.16	0.16	0.10	0.10	0.10
Top-k	0.16	0.23	0.33	0.16	0.23	0.31	0.10	0.17	0.27
Top-k, simulated prior	0.28	0.36	0.39	0.18	0.28	0.33	0.22	0.28	0.33
ITE	0.16	0.31	0.38	0.16	0.28	0.35	0.10	0.24	0.32
ITE, no repeat	0.16	0.31	0.41	0.16	0.29	**0.38**	0.10	0.24	0.35
ITE, joint usage	0.16	0.34	0.42	0.16	0.29	**0.38**	0.10	0.26	0.34
APROL-like	0.16	0.40	0.45	0.16	0.30	0.37	0.10	0.31	0.38
Robust TD3	**0.46**	**0.46**	0.46	**0.27**	0.27	0.27	**0.34**	0.34	0.34
ITE-SD (ours)	0.25	0.41	0.45	0.17	0.31	0.37	0.19	0.34	0.39
ITE-Sig (ours)	0.25	**0.46**	**0.48**	0.17	**0.33**	0.37	0.19	**0.37**	**0.40**

differences of at most 0.02. The final banks were computed on a different GPU and JAX version than in a pilot run; on the 9 repertoires computed twice, the per-gait means agree with correlation 0.976 and differ by 0.031 on average, the same order as the difference between two halves of the episodes of a single bank (0.030).

7.2 In-family damages (H2, H5)

On in-family damages, ITE-SD beats ITE after 3, 5 and 10 trials (after 5: +0.079, better on 250 units and worse on 108, $p < 0.001$) and top-k (+0.128, $p < 0.001$). Its first trial alone is better than one trial of ITE (0.25 against 0.16), since the simulated prior mean already moves the robust gaits up. The gain holds for every robot (after 5 trials: Ant +0.035, Walker2d +0.183, HalfCheetah +0.066, all $p \leq 0.001$) and every QD algorithm (MAP-Elites +0.099, PGA-ME +0.060, DCRL-ME +0.079): H2 holds and its part of H5 too, with the caveat of Section 7.3 on what the gain means on Walker2d. The part of H5 about H1 does not (Section 5). The prior selection of APROL is a strong baseline (0.40 after 5 trials, against 0.41 for ITE-SD and 0.31 for ITE); the ablations (Table 4) show that most of the gain of ITE-SD comes from its prior mean, and that 16 simulated damages already give most of it.

7.3 Does the gain mean walking further?

The return of a gait is not only how far it goes: it also counts control costs and, for Ant and Walker2d, a bonus for every step the robot stays up, and u is measured from a robot that plays no action. To

Table 3: IQM of u after 5 trials per robot and condition family (ID: in-family damages, OOD: out-of-family damages, Gap: reality gap with or without damage).

	Ant			Walker2d			HalfCheetah		
	ID	OOD	Gap	ID	OOD	Gap	ID	OOD	Gap
Top-k	0.32	0.23	0.20	0.02	-0.03	0.06	0.35	0.52	0.23
ITE	0.38	0.28	0.28	0.11	0.04	0.14	0.36	**0.53**	0.24
APROL-like	0.41	0.28	0.30	0.37	0.12	0.34	0.43	**0.53**	0.30
Robust TD3	**0.84**	**0.34**	**0.69**	0.06	-0.02	0.07	**0.52**	0.46	0.30
ITE-SD (ours)	0.44	0.29	0.34	0.36	0.13	0.38	0.44	0.51	0.30
ITE-Sig (ours)	0.48	0.32	0.37	**0.42**	**0.14**	**0.43**	0.47	0.51	**0.34**

Table 4: Ablations, IQM of u after 3 and 5 trials, all robots pooled.

	In-family damages		Out-of-family damages		Reality gap (+ damage)	
Trials	3	5	3	5	3	5
ITE-SD (ours)	0.39	0.41	0.27	0.31	0.31	0.34
ITE-SD, intact mean	0.26	0.31	0.22	0.27	0.19	0.24
ITE-SD, $N = 4$	0.31	0.34	0.23	0.28	0.25	0.28
ITE-SD, $N = 16$	0.37	0.40	0.26	0.30	0.29	0.32
ITE-SD + Matérn	0.40	0.42	0.28	0.31	0.31	0.34
ITE-Sig (ours)	**0.44**	**0.46**	**0.29**	**0.33**	**0.35**	**0.37**
ITE-Sig, no check	**0.44**	**0.46**	**0.29**	**0.33**	**0.35**	**0.37**

check what the gains in u mean, the gaits recommended after 5 trials by ITE, ITE-SD and ITE-Sig (first repetition of the replay, every final repertoire and in-family damage) were played again on the damaged robot, and their distance compared with that of the best intact gait on the intact robot.

On HalfCheetah, the learned models do recover more locomotion: the recommended gaits cover 49 % of the intact distance with ITE, 54 % with ITE-SD and 57 % with ITE-Sig, and go further than ITE's in 57 % and 60 % of the 90 (repertoire, damage) pairs (paired Wilcoxon on distances, $p = 0.004$ and $p = 0.003$). On Ant, they cover 45 %, 53 % and 55 % of the intact distance, and go further than ITE's in 64 % and 64 % of the 180 pairs ($p < 0.001$ and $p < 0.001$). On these two robots, the gains in u are gains in locomotion. On Walker2d, a robot that plays no action falls, so a gait that stands without falling already has $u > 0$, and this is what the learned prior exploits: 39 % of the gaits chosen by ITE-SD and 47 % of those chosen by ITE-Sig stay up for the whole episode, against 20 % for ITE, but they do not walk further (18 % and 20 % of the intact distance, against 25 % for ITE; $p = 0.298$ and $p = 0.858$). On Walker2d, the learned model has found that standing gaits survive damages, which the return rewards; no method recovers walking.

7.4 Beyond the simulated family (H3)

Out of the family, ITE-SD is not worse than ITE but no longer clearly better: +0.010 after 5 trials ($p = 0.089$); the hybrid kernel does not change this (Table 4). Under a reality gap, alone or combined with a damage, the gain remains (+0.072, $p < 0.001$). The damage-robust policy degrades out of its family, as H3 also predicted (Section 8). H3 is supported, with a non-inferiority that we observe but did not test formally. On HalfCheetah, out-of-family damages leave the intact ranking almost valid and no method beats top-k (0.52).

7.5 Proprioceptive signatures (exploratory)

Reading the trials through proprioception improves ITE-SD in every condition family (Figure 10 shows one adaptation). After 5 trials, ITE-Sig adds +0.034 in the family ($p < 0.001$), +0.025 under a reality gap ($p < 0.001$) and +0.017 out of the family after 3 trials ($p = 0.019$; after 5, $p = 0.131$). Against ITE, the gain after 5 trials is +0.114, +0.027 and +0.097. It holds on the three robots (Table 3). The familiarity check, on the other hand, is inert: with the selected weight, ITE-Sig and its version without the check differ on 9 + 6 out-of-family units only. A small weight already makes signatures harmless when they are misleading.

7.6 When does a model help? (H4)

The gain of ITE-SD over top-k after 5 trials is larger when the intact ranking is a poor guide (Figure 5, right): Spearman correlation between the gain and the intact/damaged rank correlation −0.38 in the family ($p < 0.001$), −0.23 out of it and −0.34 under a reality gap. The share of the response that lies in the span of the simulated responses does not predict the gain (−0.04, $p = 0.480$ in the family). H4 is supported for the fidelity of the prior and not for the span: when the prior is right, there is little to gain, whatever the model.

8 Repertoires or reinforcement learning?

A repertoire is not the only way to recover from damage. Reinforcement learning (RL) can re-train a policy on the damaged robot, or train in advance a single policy that tolerates damages. We compare both with adaptation, counting episodes on the damaged robot.

8.1 Re-training on the damaged robot

TD3 [13] is trained on the damaged Ant, from scratch or from the best intact DCRL-ME policy (same architecture, new critic), on 4 damages (FR leg paralysed, HL leg weakened, FL+FR legs paralysed, FL ankle paralysed) and 2 seeds, for 2000 episodes with 16 robots in parallel and one gradient step per transition; the policy is evaluated on 8 episodes after every 16 training episodes. Since the new critic is untrained at first, we also test two remedies: a *warm-up*, where the actor is frozen during the first 160 episodes on the damaged robot (counted), and a *pre-trained critic*, trained for 512 episodes on the intact robot in simulation with the actor frozen (not counted, since no damaged robot is involved).

Within 2000 episodes, TD3 from scratch matches the gait found by ITE in 3/8 runs (median 1264 episodes when it does), and plain fine-tuning in only 2/8 runs, one of them because the intact policy already matched ITE before any training (Figure 11 and Table 18). Plain fine-tuning collapses at the first evaluation, because it follows the gradient of an untrained critic: with a critic warm-up it matches ITE in 8/8 runs (median 448 episodes), and with a pre-trained critic in 8/8 runs (median 400 episodes). Even so, the best re-training needs hundreds of episodes on the damaged robot, against 1 to 20 trials for a repertoire.

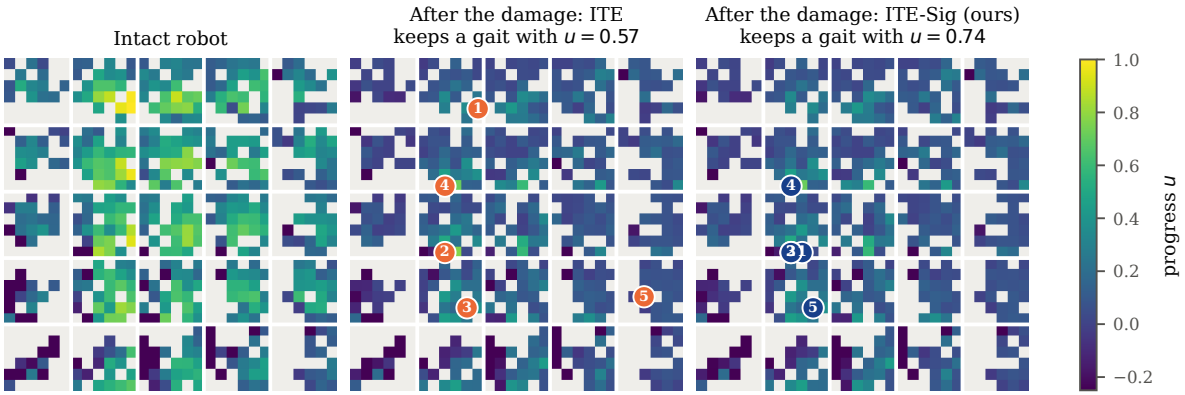


Figure 10: The five gaits tried by ITE (orange) and ITE-Sig (dark blue) after leg 2 of the Ant is paralysed (DCRL-ME repertoire, seed 0; the repetition closest to the average of each method). Colour is the progress of each gait, intact (left) and damaged (middle, right). ITE starts from the best intact gait and explores around it in descriptor space; ITE-Sig starts from gaits that resist the simulated damages, refines its estimate of the damage with each trial, and keeps a better gait.

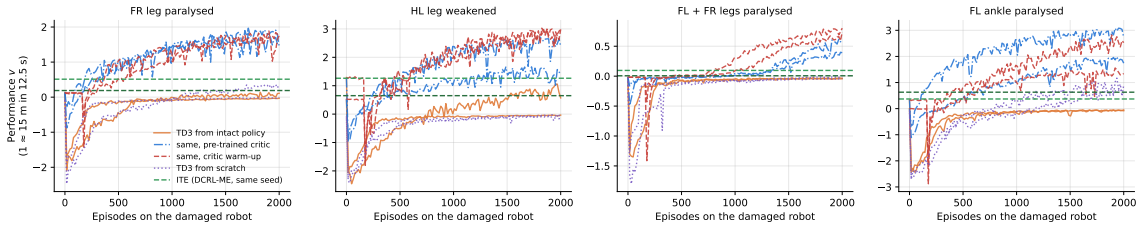


Figure 11: Ant: TD3 on the damaged robot (two seeds each), from scratch, from the best intact DCRL-ME policy, and from the same policy with a critic warm-up or a pre-trained critic, against the gait found by ITE on the DCRL-ME repertoire of the same seed (dashed lines).

8.2 A damage-robust policy

The strongest alternative to adaptation is not to adapt at all: train one policy that works under many damages. We train TD3 from scratch in simulation on random damages of the in-family distribution (Section 3.2), with a new damage at every episode, for 10,000 Ant episodes and the same number of simulated transitions (2.5 million) on the other robots, 2 seeds per robot, and evaluate it without any trial (32 episodes per condition). The in-family test damages lie inside its training distribution, which favours it: unlike ITE, it needs to know in advance what kind of damage can happen.

On Ant, with fresh episodes. Without any trial, the robust policy reaches a median $v = 1.46$ over the damages, better than ITE on DCRL-ME repertoires in 10 of the 12 damages and as good as ITE on the best repertoires, those of PGA-ME (better in 6 of 12; median $v = 1.51$ for ITE; Table 19). It is a strong intact gait ($v = 2.06$) and is little affected by the reality gap ($v = 1.40$ against 1.24 for ITE on PGA-ME). It fails, however, when two legs are paralysed, where it does worse than a standing robot, while the repertoires still offer a gait that barely moves.

On three robots, against the learned models. On Ant it is the strongest method within its training family (0.84, against 0.48 for ITE-Sig after 5 trials): it was trained on this very family of damages, needs no trial, and is a much faster gait to begin with: on the intact Ant it reaches $u = 1.90$, i.e. well above the best gait of the repertoires, whose QD algorithms optimise diversity as well as speed. Out of the family, its advantage on Ant shrinks to 0.34 against 0.32, and on HalfCheetah it is below ITE-Sig under a reality gap (0.30 against 0.34). On Walker2d, the robust policy did not learn to walk: with the same number of simulated transitions as on Ant (2.5 million), TD3 stayed in the local optimum of standing, which the alive bonus of Walker2d rewards, and reaches only $u = 0.12$ on the intact robot. We report its Walker2d numbers for completeness but draw no conclusion from them. The comparison on Ant and HalfCheetah suggests that the two approaches are complementary: a robust policy as the first gait, a repertoire for the trials that follow.

9 Discussion

The model is the bottleneck. The story of this work is that of a negative result turned into a diagnosis. On Ant, ITE was barely better than a ranking (Section 5), and a descriptor that models the damage three times better did not change this. Yet the repertoires keep good gaits after damage (Section 4), these gaits are often far down the intact ranking, and damages act along a few directions. With the same acquisition, changing only the prior and the kernel

raises the share of the intact progress recovered after three in-family trials from 0.26 (ITE) to 0.39 (ITE-SD) and 0.44 (ITE-Sig). The behaviour descriptor remains essential to *build* a diverse repertoire, but it should not be assumed to be the right similarity to *search* it after a damage. A low-dimensional description of how gaits react to damages can be learned cheaply from the simulator and is a more useful "adaptation descriptor".

The quality of the repertoire still matters most. What a damaged robot finally gets depends first on the repertoire: PGA-ME and DCRL-ME repertoires are as resilient as MAP-Elites ones in relative terms and keep much better gaits in absolute terms. Re-evaluating the elites before deployment, offline in simulation, is cheap and necessary, since MAP-Elites keeps the luckiest evaluations; this is the problem studied by uncertain QD [11].

What the simulated family does not cover. Out of the family, the learned model does not hurt but no longer helps much, and on HalfCheetah no method beats the intact ranking. The learned kernel is only as good as the damage family; the hybrid kernel did not fix this. Proprioceptive signatures help more there, and a richer family of simulated damages is the most direct next step. The familiarity check of ITE-Sig never triggered usefully: with the selected weight, signatures are already used with caution.

Repertoires and RL. RL re-training needs hundreds of episodes on the damaged robot, far more than a repertoire. A damage-robust policy needs none and, on Ant, is better than any adaptation within its family; it loses this advantage under damages it was not trained for, and two paralysed legs defeat it. On Walker2d our TD3 baseline failed to learn a gait, so the comparison there is not informative. Since a robust policy can be the first gait to try and a repertoire offers the alternatives, combining them is a natural extension.

Counting episodes, not compute. In simulation, RL consumes thousands of episodes in minutes; on a real robot, each episode costs time, wear and supervision. We therefore compare trials and episodes, not wall-clock time.

10 Limitations

Everything happens in simulation: a trial is a simulated episode, and the reality gap is itself simulated by changing a few physical parameters. Damages are modelled as changes of motor strength (in family) or of a few other quantities (out of family). One repertoire size and one budget are used, with the hyper-parameters of the official notebooks, and 3 to 5 seeds per algorithm, so that the (repertoire, condition) units are not independent; mixed models account for this only approximately. Replaying trials from stored episodes is validated against fresh episodes on Ant only. On Walker2d, the return rewards standing as much as walking, so the gains there mean not falling rather than walking again. The signatures assume access to the simulated sensors, with only episode-to-episode noise. ITE-Sig was designed after the hypotheses were fixed; its settings were chosen on separate development repertoires, but its results are exploratory. RL re-training was tested on 4 damages and 2 seeds; the robust TD3 baseline failed on Walker2d; no meta-learned or history-based adaptive policy [16, 18] was tested. Finally, H1 holds only at 20 trials and not on every robot, and the span of the simulated responses, which we expected to explain the gains, does not.

11 Conclusion

Repertoire-based damage recovery is limited less by the search than by the model that guides it. Modern QD algorithms build repertoires that remain useful after damage, and adaptation from them needs a handful of trials where RL re-training needs hundreds of episodes. But with the model of ITE, the Bayesian search adds little to a ranking. Learning the model from a few dozen simulated damages, and reading the robot's proprioception during trials, makes it worthwhile on three robots and three QD algorithms. Two questions remain open: the damage the simulator did not anticipate, and how to combine a repertoire with a damage-robust policy. Replaying trials from condition banks makes the number of trials cheap to measure; we hope it becomes a standard measure of adaptation again.

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A Additional results of the Ant experiments

This appendix gives the detailed tables of the experiments with fresh episodes on Ant (Sections 4, 5 and 8).

Oracle bias. Re-evaluating the cell chosen by the oracle on 16 fresh episodes shows that the oracle over-estimates the best gait by $v \approx 0.06$ (MAP-Elites), 0.08 (PGA-ME) and 0.06 (DCRL-ME) on average (Table 12). With these unbiased values the median resilience is 50 %, 66 % and 60 %; the mixed model now finds PGA-ME slightly more resilient than MAP-Elites (difference +0.08 [+0.01, +0.16] with the correction, $p = 0.030$, against +0.04 [−0.04, +0.12] without it), but the rank test does not. Either way, building the repertoire with RL does not make it less resilient.

Mixed models. For comparisons between algorithms we fit `value ~ algorithm + damage + (1 | repertoire)`, and for paired comparisons between methods `difference ~ algorithm + (1 | repertoire)`, and report differences with 95 % confidence intervals; with 3 repertoires per algorithm these intervals are approximate. They confirm the comparisons between algorithms (Table 13): no significant difference in relative resilience, large differences in absolute performance. For ITE against top-k (Table 14) they agree with the Wilcoxon tests on MAP-Elites and PGA-ME, but on DCRL-ME find a positive mean gain after three and five trials while the median gain is zero. The pooled medians of Table 10 suggest a larger gap on DCRL-ME repertoires (0.59 against 0.48 after ten trials), but the paired test does not support it ($p = 0.407$). On PGA-ME repertoires, the best intact gait already keeps most of its performance ($v = 1.43$ at the first trial); on MAP-Elites repertoires, all methods stay close to a standing robot.

Reality gap. With the gap, ITE's lead over top-k after three trials is +0.16 on DCRL-ME ($p = 0.003$), and on PGA-ME top-k is ahead after ten trials (−0.08, $p = 0.009$). A moderate gap makes the prior less reliable without changing the conclusion: the model mostly helps in the first trials. The joint-usage runs were made on a CPU; since GPU and CPU results differ slightly for the same seed, ITE and top-k were re-run on the same CPU for these comparisons.

Continuous descriptor space. ITE on the actor of DCRL-ME is better than ITE on the cells in only 3 of the 36 (run, damage) pairs (Wilcoxon $p < 0.001$; 3 dedicated runs where the actor was saved). The actor is a useful source of offspring during training, but it does not replace the elites that were selected and refined.

Table 5: Main hyper-parameters (full configuration in `configs/`).

	MAP-Elites	PGA-ME	DCRL-ME
Offspring per iteration	256 mutations	128 mutations + 128 PG	128 + 64 PG + 64 actor
Iso/line variation σ	0.005 / 0.05	0.005 / 0.05	0.005 / 0.05
Critic	–	TD3, 256 × 256	descriptor-cond., 256 × 256
Critic steps / iteration	–	300	3000
PG steps / iteration	–	100	150

Table 6: Repertoires at the end of training (median over seeds).

Algorithm	Coverage (%)	Max fitness	QD-score ($\times 10^3$)
MAP-Elites	72.3	1235	814
PGA-ME	76.8	1922	962
DCRL-ME	83.8	1625	1074

Table 7: Resilience (%) and absolute oracle performance v after damage (median over seeds).

Damage	MAP-Elites (%)	PGA-ME (%)	DCRL-ME (%)	MAP-Elites (v)	PGA-ME (v)	DCRL-ME (v)
FL ankle paralysed	54	74	48	0.28	1.86	0.83
HR ankle paralysed	82	81	82	0.35	2.06	1.48
FL leg paralysed	41	29	41	0.21	0.73	0.73
FL leg weakened	88	87	76	0.38	2.20	1.32
HL leg paralysed	32	55	51	0.13	1.72	0.92
HL leg weakened	55	85	74	0.24	2.33	1.32
HR leg paralysed	61	58	65	0.23	1.79	1.18
HR leg weakened	69	83	87	0.28	2.15	1.56
FR leg paralysed	46	31	30	0.20	0.97	0.54
FR leg weakened	91	89	81	0.40	2.25	1.42
FL + HL legs paralysed	11	4	11	0.06	0.10	0.19
FL + FR legs paralysed	23	5	10	0.12	0.16	0.18

Table 8: Share of the intact progress (%) recovered after a given number of trials (median).

Method	1 trial	3 trials	5 trials	10 trials	20 trials
ITE	18	28	30	33	33
top-k (no GP)	18	27	30	31	35
random	1	9	16	26	33

Table 9: ITE against top-k at equal numbers of trials: mean difference in absolute performance v (ITE − top-k), number of units where ITE is better / equal / worse, and paired Wilcoxon test, over the (repertoire, damage) units.

Repertoire	3 trials	5 trials	10 trials
MAP-Elites	+0.01 (18/0/18; $p = 0.581$)	+0.00 (16/0/20; $p = 0.798$)	+0.01 (20/0/16; $p = 0.451$)
PGA-ME	+0.03 (19/4/13; $p = 0.135$)	−0.03 (18/1/17; $p = 0.974$)	−0.03 (18/0/18; $p = 0.944$)
DCRL-ME	+0.13 (20/1/15; $p = 0.019$)	+0.08 (17/4/15; $p = 0.166$)	+0.04 (17/2/17; $p = 0.407$)

Table 10: Absolute performance v of the recommended gait at equal numbers of trials, by repertoire (median).

Repertoire – method	1 trial	3 trials	5 trials	10 trials	20 trials
MAP-Elites – ITE	0.07	0.10	0.11	0.11	0.12
MAP-Elites – top-k	0.07	0.09	0.10	0.09	0.11
MAP-Elites – random	-0.03	0.02	0.07	0.09	0.12
PGA-ME – ITE	1.43	1.47	1.47	1.47	1.47
PGA-ME – top-k	1.43	1.47	1.52	1.52	1.57
PGA-ME – random	0.10	0.21	0.47	0.78	1.11
DCRL-ME – ITE	0.22	0.47	0.47	0.59	0.60
DCRL-ME – top-k	0.22	0.36	0.37	0.48	0.59
DCRL-ME – random	0.03	0.22	0.28	0.49	0.59

Table 11: ITE on each kind of repertoire (median over seeds, damages and repetitions).

Repertoire	Trials	% intact	Absolute v
MAP-Elites	6	24	0.11
PGA-ME	3	55	1.47
DCRL-ME	6	33	0.59

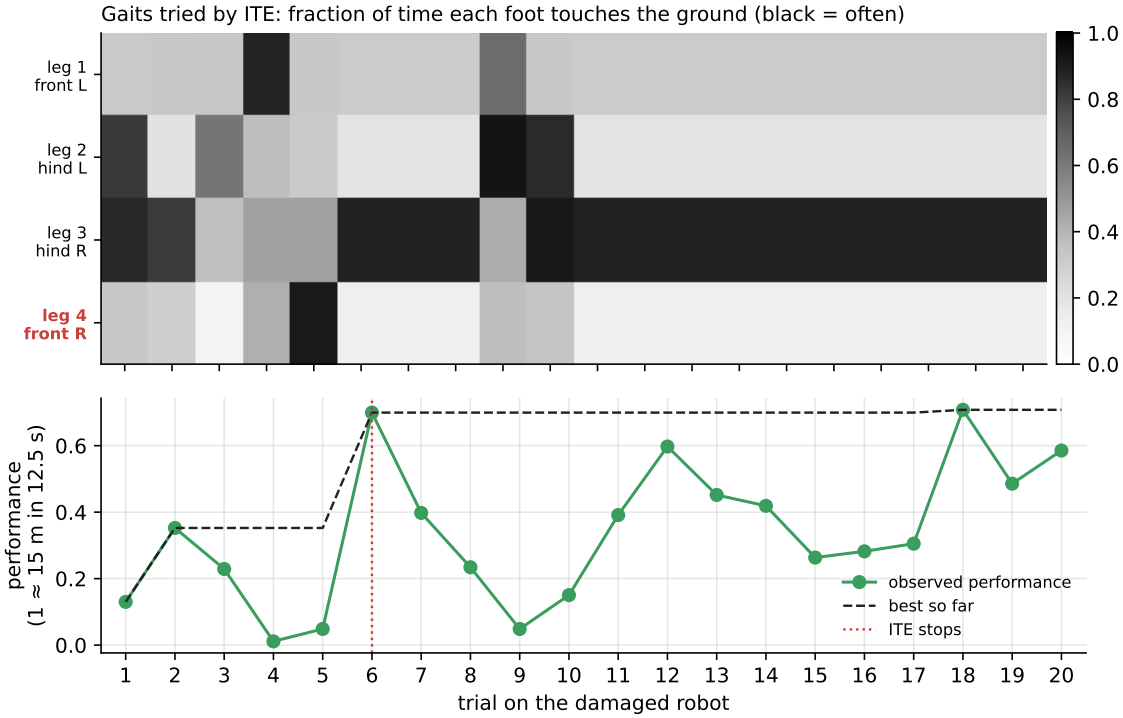
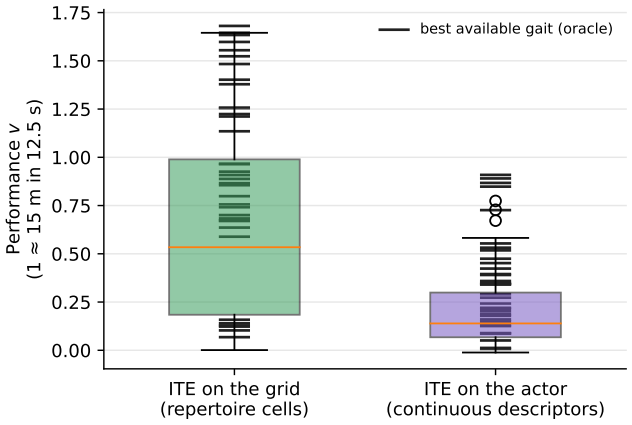


Figure 12: One ITE run on a DCRL-ME repertoire (front-right leg paralysed; the five repetitions all recommend the same gait, after 6 or 7 trials, and this one used 6): feet-contact profile of each gait tried, and its observed performance.

Table 12: Oracle re-evaluated on fresh episodes: mean bias of the 4-episode oracle, and medians of the resilience and of the best gait after damage computed with the re-evaluated values.

Repertoire	Bias of the oracle (v)	Resilience, checked (%)	Best gait after damage, checked (v)
MAP-Elites	0.06	50	0.17
PGA-ME	0.08	66	1.68
DCRL-ME	0.06	60	1.04



ITE search space	Trials	Found v	Oracle v
Grid (repertoire cells)	6	0.53	0.90
Continuous (conditioned actor)	3	0.14	0.32

Figure 13: ITE on the cells of a DCRL-ME repertoire against ITE on 2048 gaits generated by its descriptor-conditioned actor for targets spread over $[0, 1]^4$ (3 dedicated runs, same runs for both).

Table 13: Differences between algorithms from linear mixed models (fixed effects: algorithm and damage; random intercept per repertoire), with 95% confidence intervals.

Quantity	Comparison	Difference [95% CI]	p-value
Resilience	PGA-ME − MAP-Elites	+0.04 [−0.04, +0.12]	0.357
Resilience	DCRL-ME − MAP-Elites	+0.01 [−0.08, +0.09]	0.888
Resilience	DCRL-ME − PGA-ME	−0.03 [−0.11, +0.05]	0.435
Best gait after damage (v)	PGA-ME − MAP-Elites	+1.31 [+1.10, +1.51]	<0.001
Best gait after damage (v)	DCRL-ME − MAP-Elites	+0.74 [+0.53, +0.94]	<0.001
Best gait after damage (v)	DCRL-ME − PGA-ME	−0.57 [−0.78, −0.37]	<0.001
Gait found by ITE (v)	PGA-ME − MAP-Elites	+1.16 [+0.97, +1.36]	<0.001
Gait found by ITE (v)	DCRL-ME − MAP-Elites	+0.48 [+0.29, +0.68]	<0.001
Gait found by ITE (v)	DCRL-ME − PGA-ME	−0.68 [−0.88, −0.48]	<0.001

Table 14: Mean gain of ITE over top-k in v (95% confidence interval) from a mixed model per repertoire type.

Repertoire	3 trials	5 trials	10 trials
MAP-Elites	+0.01 [−0.04, +0.06]	+0.00 [−0.06, +0.07]	+0.01 [−0.07, +0.09]
PGA-ME	+0.03 [−0.02, +0.08]	−0.03 [−0.09, +0.04]	−0.03 [−0.11, +0.05]
DCRL-ME	+0.13 [+0.08, +0.18]	+0.08 [+0.02, +0.15]	+0.04 [−0.04, +0.12]

Table 15: Gaussian-process hyper-parameters fitted by maximum marginal likelihood on the development repertoires for each descriptor, and log-likelihood gained over independent cells, per (repertoire, damage) pair.

Descriptor	Length-scale	Signal var.	Noise var.	Gain per pair
feet contact (4D)	0.4	0.050	0.020	51.7
joint usage (8D)	0.2	0.100	0.003	152.2

Table 16: ITE on joint usage minus ITE on feet contact: mean difference in v and paired Wilcoxon test, over the (repertoire, damage) units.

Repertoire	3 trials	5 trials	10 trials
MAP-Elites	−0.01 ($p = 0.023$)	−0.01 ($p = 0.369$)	−0.01 ($p = 0.062$)
PGA-ME	+0.06 ($p = 0.445$)	+0.03 ($p = 0.808$)	+0.03 ($p = 0.793$)
DCRL-ME	+0.01 ($p = 0.728$)	+0.04 ($p = 0.351$)	+0.06 ($p = 0.265$)

Table 17: Adaptation with a reality gap: prior from the nominal simulator, trials on the perturbed robot (medians over repertoires, damages and repetitions; "% intact": share of the progress of the best intact gait of the perturbed robot).

Method	Trials	Nominal v	With reality gap v	% intact (gap)
best-intact	1	0.19	0.10	12
top-k	20	0.40	0.27	33
ITE	6	0.37	0.28	26

Table 18: Episodes on the damaged robot needed by each TD3 variant to match the gait found by ITE (DCRL-ME repertoire of the same seed, mean number of ITE trials in the first column); "–": not reached within 2000 episodes.

Damage	ITE trials	Scratch	Fine-tune	Warm-up	Pre-trained critic
FL ankle paralysed, seed 0	10.0	1072	–	496	112
FL ankle paralysed, seed 1	5.6	1808	–	400	576
HL leg weakened, seed 0	1.4	–	0	0	1088
HL leg weakened, seed 1	6.4	–	1536	336	224
FR leg paralysed, seed 0	6.6	–	–	496	192
FR leg paralysed, seed 1	15.0	1264	–	224	192
FL + FR legs paralysed, seed 0	18.8	–	–	1008	1344
FL + FR legs paralysed, seed 1	20.0	–	–	720	720

Table 19: Performance v after damage of the damage-robust TD3 policy (zero trials, mean of 2 seeds) and of the gait found by ITE (mean over seeds and repetitions), on the nominal robot and with the reality gap of Section 5.

Damage	Robust TD3	ITE, DCRL-ME	ITE, PGA-ME	Robust TD3 (gap)	ITE, PGA-ME (gap)
FL ankle paralysed	1.29	0.50	1.35	1.38	1.05
HR ankle paralysed	2.21	1.09	2.05	1.99	1.64
FL leg paralysed	0.55	0.39	0.48	0.73	0.32
FL leg weakened	2.09	0.87	1.96	1.91	1.67
HL leg paralysed	0.75	0.35	1.66	0.48	1.44
HL leg weakened	1.89	0.86	2.26	1.75	2.00
HR leg paralysed	1.63	0.59	0.91	1.43	0.87
HR leg weakened	2.25	1.13	1.90	2.08	1.64
FR leg paralysed	0.63	0.36	0.71	0.48	0.63
FR leg weakened	2.26	1.20	2.25	1.99	1.92
FL + HL legs paralysed	-0.14	0.06	0.03	-0.21	0.01
FL + FR legs paralysed	-0.48	0.03	0.04	-0.36	0.06

B The learned model in plain words

The Ant experiments ended on a puzzle: ITE was barely better than trying the best intact gaits in order. Is the Bayesian search useless with modern repertoires, or is it given the wrong information? This appendix summarises, without formulas, how Sections 6 to 8 answer that question.

B.1 Idea

ITE needs two pieces of knowledge before the first trial. The *prior* says how good each gait is expected to be after the damage; ITE uses the intact performance. The *kernel* says which gaits react alike, so that one trial also teaches something about other gaits; ITE assumes that gaits with close behaviour descriptors (here, how often each foot touches the ground) react alike. Neither is learned from damages.

Yet the simulator that built the repertoire can also break it. Before deployment, we play every gait under $N = 64$ random *simulated damages* (each motor paralysed or weakened with probability 0.25), which costs a few minutes of GPU and no trial on the robot. For each gait i and simulated damage k, this gives the change of performance R_{ki}. Two numbers then come out directly:

- the *mean response* of each gait over the simulated damages, added to the intact performance, is a prior that already favours gaits that usually survive damages;
- the *covariance* of the responses of two gaits (do they tend to lose together, or not?) is a kernel that says which gaits react alike, whatever their descriptors.

ITE with this prior and this kernel, and without retrying a gait, is called *ITE-SD* (ITE with Simulated Damages). Mathematically, it assumes that the real damage is an unknown combination of the simulated ones: each trial measures this combination, and the model then predicts all gaits at once.

ITE-Sig goes one step further. During a trial, the robot does not only measure how far it went: it also records its joint angles and speeds, the motion of its torso and its foot contacts. We summarise this *proprioceptive signature* by a few numbers per episode. The simulated damages tell how each gait's signature changes under each damage, so a trial now gives many measurements of the damage instead of one. ITE-Sig also checks whether the signatures look like those of the simulated damages and ignores them otherwise.

B.2 Setup

- **Robots and repertoires.** Ant, Walker2d (two legs, falls end the episode) and HalfCheetah, with MAP-Elites, PGA-ME and DCRL-ME at equal budget: 5 seeds per algorithm for Ant, 3 for the others, 33 repertoires. Shorter development runs are used only to choose the settings of every method, which are then frozen.
- **Conditions.** Every gait is played under the intact robot, the 64 simulated damages, and the test conditions: *in-family* damages (paralysed or weakened limbs, like the simulated ones but drawn separately), *out-of-family* damages that act differently (a motor stuck pushing, a motor wired backwards, the sensors of a leg reading zero, a slippery or a heavy foot), and a *reality gap* (friction, mass and motor strength changed at two levels, with or without a damage). In total 63 test conditions for Ant.
- **Replay.** Since every gait was played under every condition, adaptation is replayed from these stored episodes: a trial picks one stored episode of the chosen gait, and the recommended gait is scored on other episodes that the method never sees. This makes each method cheap enough to repeat 20 times per (repertoire, condition) pair. Replayed ITE and top-k match runs with fresh episodes on Ant (correlation 0.97–0.99).
- **Methods.** Top-k, ITE, ITE without repeats, top-k on the simulated prior, an APROL-like method that picks, after each trial, the simulated damage whose prior best explains the trials [15], ITE-SD, ITE-Sig, and the damage-robust TD3 policy, which plays without any trial.
- **Measure.** u = share of the progress of the best intact gait recovered by the recommended gait ($u = 1$: as good as before the damage, $u = 0$: as a robot standing still). Results are interquartile means (IQM: the mean of the middle half of the values, robust to outliers) with 95 % bootstrap intervals. Five hypotheses (H1–H5) were written before the final runs; ITE-Sig, designed later, is reported as exploratory.
- **Compute.** Repertoires on Kaggle; condition banks and replay on 25 machines of the university's teaching rooms (RTX 5070), which needed a newer JAX version; the recomputed banks agree with earlier ones up to episode noise.

B.3 Results

The model was the bottleneck (H2, H5). On in-family damages, ITE-SD recovers 0.41 of the intact progress after 5 trials, against 0.31 for ITE and 0.23 for top-k (Figure 9 and Table 2); the difference with ITE is +0.079 ($p < 0.001$). It holds on each robot and for each QD algorithm. On Walker2d, ITE with the foot-contact kernel recovers only 0.11, against 0.36 for ITE-SD: the descriptor that makes a repertoire diverse is a poor guide to how its gaits react to a damage. On Walker2d, however, the score rewards staying upright as much as walking: the gaits that ITE-SD picks there mostly avoid falling (39 % of them stay up the whole episode, against 20 % for ITE) but do not walk further.

ITE retries gaits (H1). Forbidding repeats helps ITE after 20 trials (+0.022, $p = 0.001$), but not significantly after 10, and only on Ant when robots are taken separately.

Out of the family (H3). For damages unlike the simulated ones, ITE-SD is not worse than ITE but no longer clearly better (+0.010, $p = 0.089$). Under a reality gap the gain remains (+0.072). On HalfCheetah, no method beats top-k out of the family.

Proprioception helps (exploratory). ITE-Sig improves on ITE-SD in every family: after 5 trials +0.034 in the family and +0.025 under a reality gap, and +0.017 out of the family after 3 trials. Its safety check never mattered: the selected weight on signatures is small enough for misleading signatures to do no harm.

When does a model help? (H4). The gain over top-k is larger when the intact ranking predicts badly the ranking after damage (Spearman correlation -0.38 in the family). We expected it also to grow when the real damage looks like a combination of simulated ones; it does not.

Robust policy. On Ant, the TD3 policy trained on random damages is better than any adaptation within its training family (0.84 against 0.48 for ITE-Sig after 5 trials); it is also a much faster gait on the intact Ant ($u = 1.90$, far above the best gait of the repertoires), and loses most of this advantage out of the family (0.34 against 0.32). On Walker2d, with the same number of simulated transitions, TD3 only learned to stand ($u = 0.12$ on the intact robot), so the comparison there is not informative.

Takeaway. The Ant experiments suggested that ITE adds little to a good ranking. The learned model shows why: with the behaviour descriptor as kernel and the intact performance as prior, ITE has little to work with. Learning both from a few dozen simulated damages makes the Bayesian search worthwhile, on three robots and three QD algorithms, and reading the robot's proprioception during trials improves it further. What remains open is the damage that the simulator did not anticipate.

C Reproducibility and compute

```
python -m qd_damage.repertoires -algo {me,pgame,dcrlme} -seed S
python -m qd_damage.oracle -runs results/repertoires/<run>
python -m qd_damage.oracle -check-best -runs results/repertoires/<run>
python -m qd_damage.adaptation -runs results/repertoires/<run>
python -m qd_damage.calibrate [-joint-usage] -oracles results/oracle/*_demo
python -m qd_damage.joint_usage -runs results/repertoires/<run>
python -m qd_damage.adaptation -joint-usage -methods ite ...
python -m qd_damage.oracle -reality-gap -out results/oracle_gap -runs ...
python -m qd_damage.adaptation -reality-gap -oracle-root results/oracle_gap ...
python -m qd_damage.rl_retrain -scenario <s> -variant <variant> -seed S
python -m qd_damage.rl_retrain -robust -seed S
python -m qd_damage.analysis.report
python -m qd_damage.bank -signatures -runs results/repertoires/<run>
python -m qd_damage.feedback -runs results/repertoires/<run>
python -m qd_damage.tune {-gp,-search} -banks results/bank/*_demo
python -m qd_damage.offline -banks results/bank/<run>
python -m qd_damage.analysis.replay_check
python -m qd_damage.analysis.gecco
```

Versions: QDax 0.5.0, JAX 0.4.28, Brax 0.10.4 (pinned; see `requirements-lock.txt`). The GPU runs were made on Kaggle (2× Tesla T4) with the scripts of `kaggle/`; the oracle check, the joint-usage descriptor and its CPU control on a laptop CPU. For the replayed experiments, the condition banks and the replay ran on the PPTI machines (RTX 5070) with JAX 0.5.3 (scripts and setup in `ppti/`). Every run stores its configuration, seed, code commit, library versions and device.

Compute of the second set of experiments: about 13.1 hours of Kaggle sessions (repertoires of Walker2d, HalfCheetah and Ant seeds 3–4), 4.2 GPU-hours on 25 RTX 5070 of the university's teaching rooms (condition banks), and a few hours of laptop and CPU time for tuning, replay and analysis (`docs/compute_budget.md`).